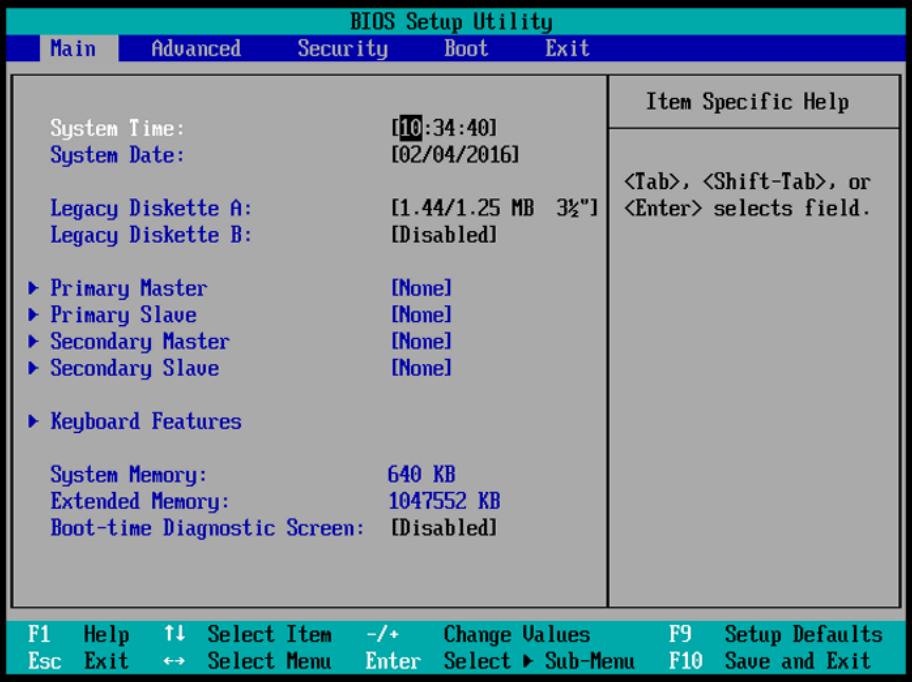
BIOS and UEFI

BIOS stands for __Basic Input Output System_____-it is software stored on a _BIOS Chip_____ on a computer that helps a computer start up



-older→sometimes referred to as ___Legacy_____ BIOS
-____text based_____/tab based interface



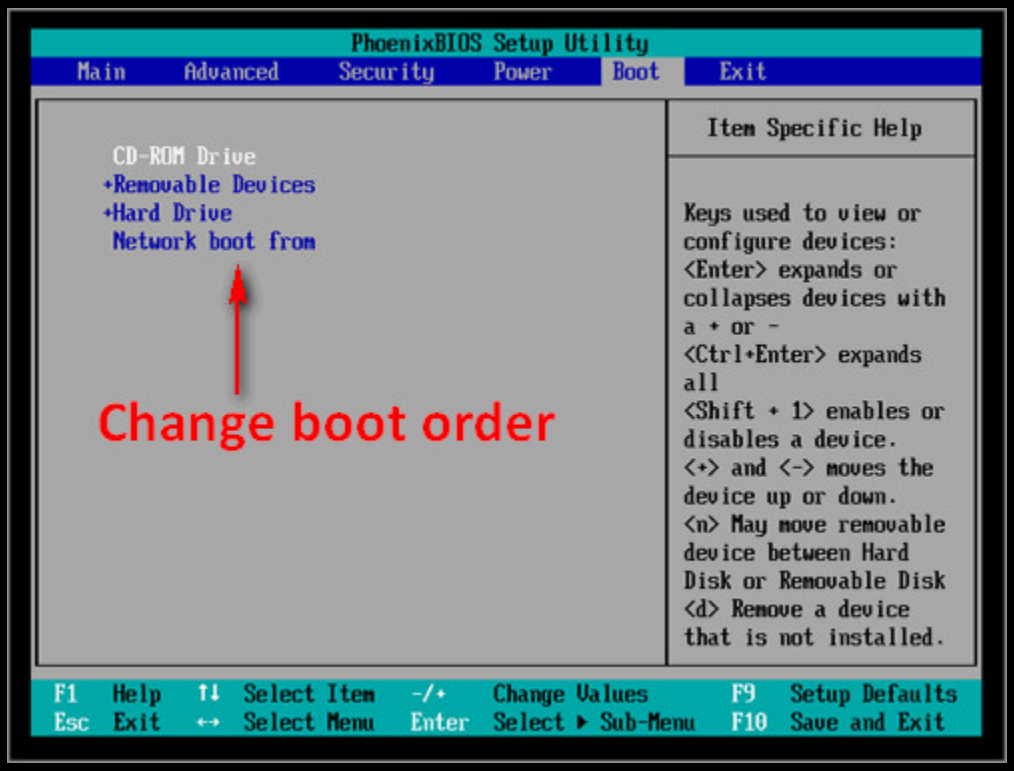
-2005→new software developed
UEFI-_Unified Extensible Firmware Interface_____
-__graphical interface_____ (mouse input) and can access 2TB drives



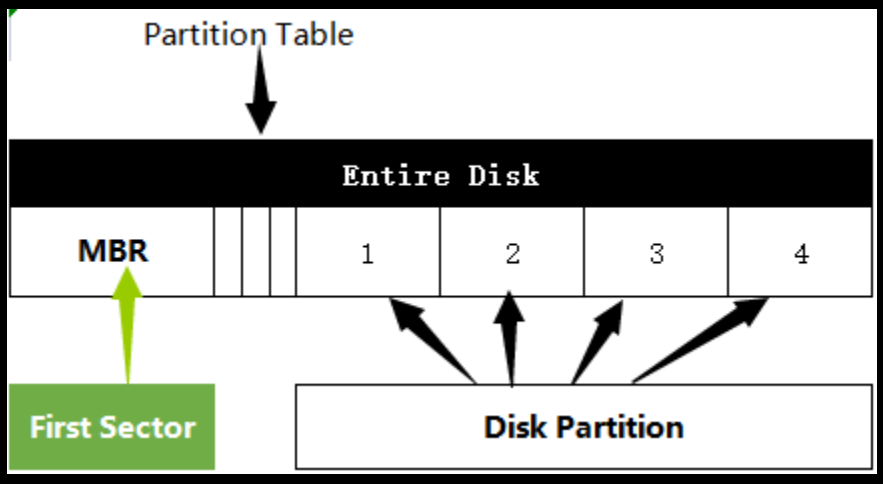
-replaces BIOS
-sometimes referred to as _modern BIOS or UEFI_____ BIOS

What does it do?

The BIOS is a chip on your MB that contains firmware (software embedded, usually ___read-only_____ , on a chip), that helps your computer to start up. One of the first things it does is to perform the POST(__Power-On Self Test_____) which is a set of procedures it follows to make sure everything is __working correctly _____ i.e. memory, HDD, peripheral devices like the keyboard and mouse etc. If not, it is responsible for alerting you through on __screen messages_____ and __beep/LED_____ patterns. If everything is working then it will find the first bootable device as listed in the CMOS settings.



It searches for the MBR(Master Boot Record_____) in this device. This is usually the first part of the storage device and contains information on where to find and run the bootloader on that device.

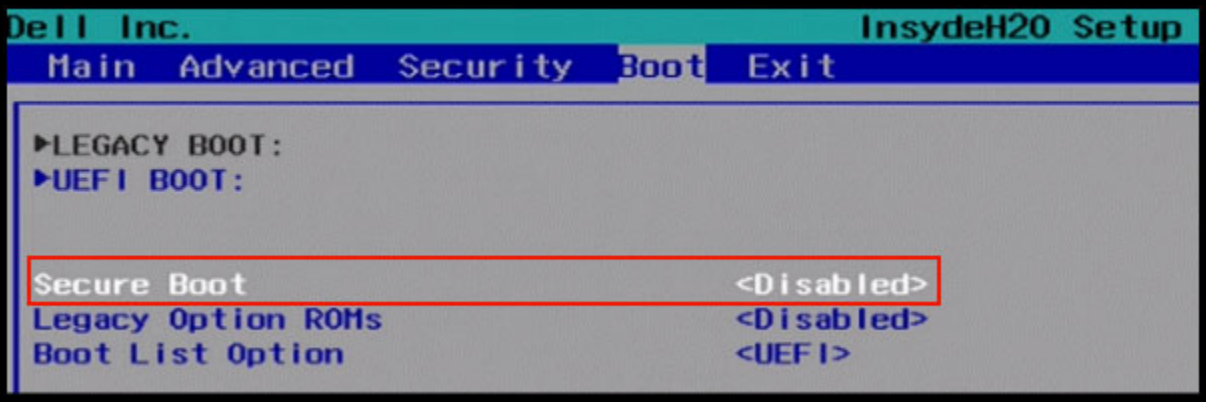


The bootloader has instructions on where to find the Operating System in the bootable device and then loads the OS into memory. The OS then takes over your computer and starts it up.

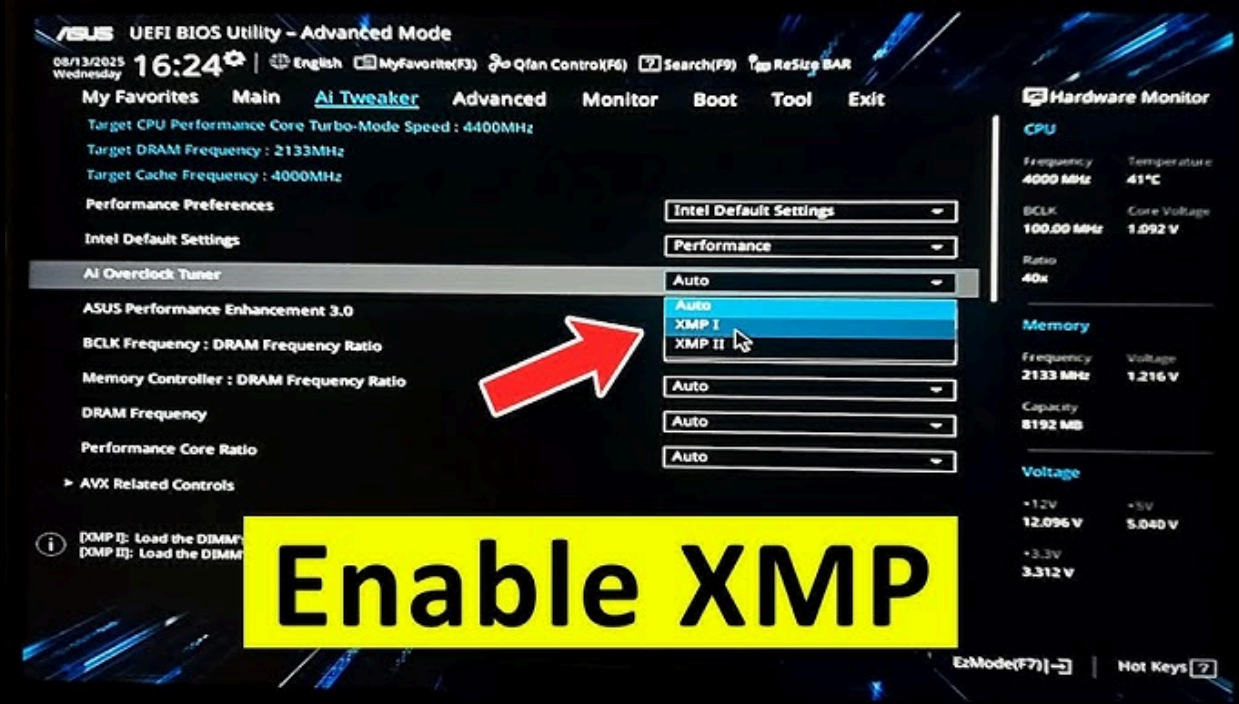
Common Settings

-access on Boot Up via Delete, Esc or F2__ _____keys (usually shown in a message on the screen during boot process)

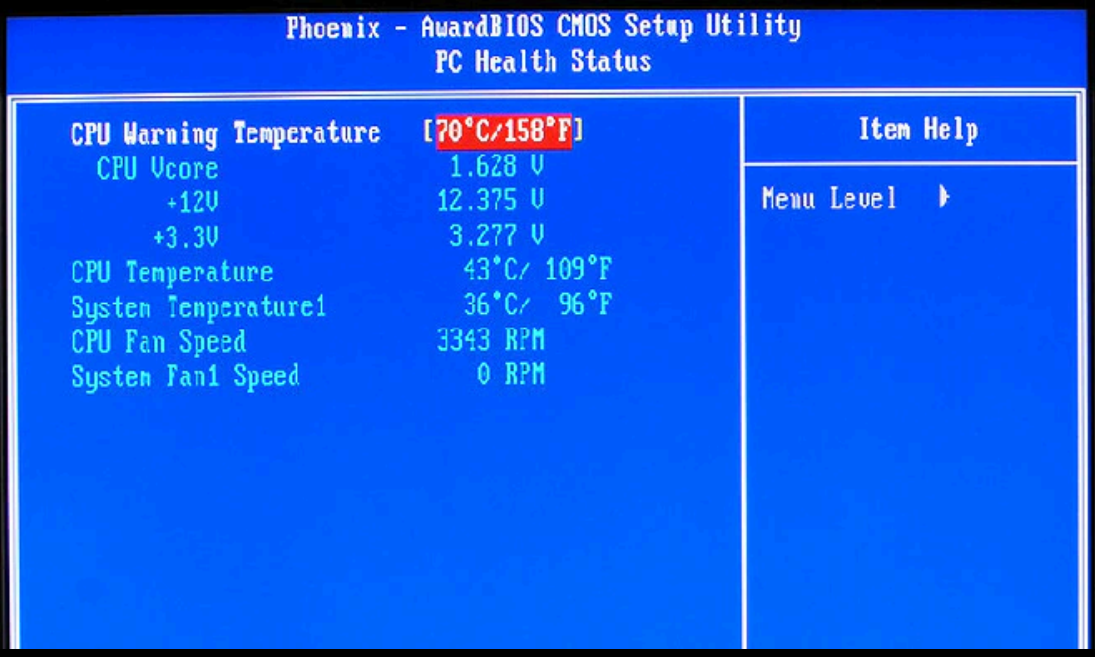
1. **Boot Options**→changes how the PC will ____boot_____up i.e. which devices will be search for an OS (the order to search)
2. **Secure Boot**→__enable_____ or ___disable_____options→came in with windows 8→when enabled it prevents any other OS other than windows from booting up (must be disabled to boot to linux)-->might be referred to as windows 8/10 options



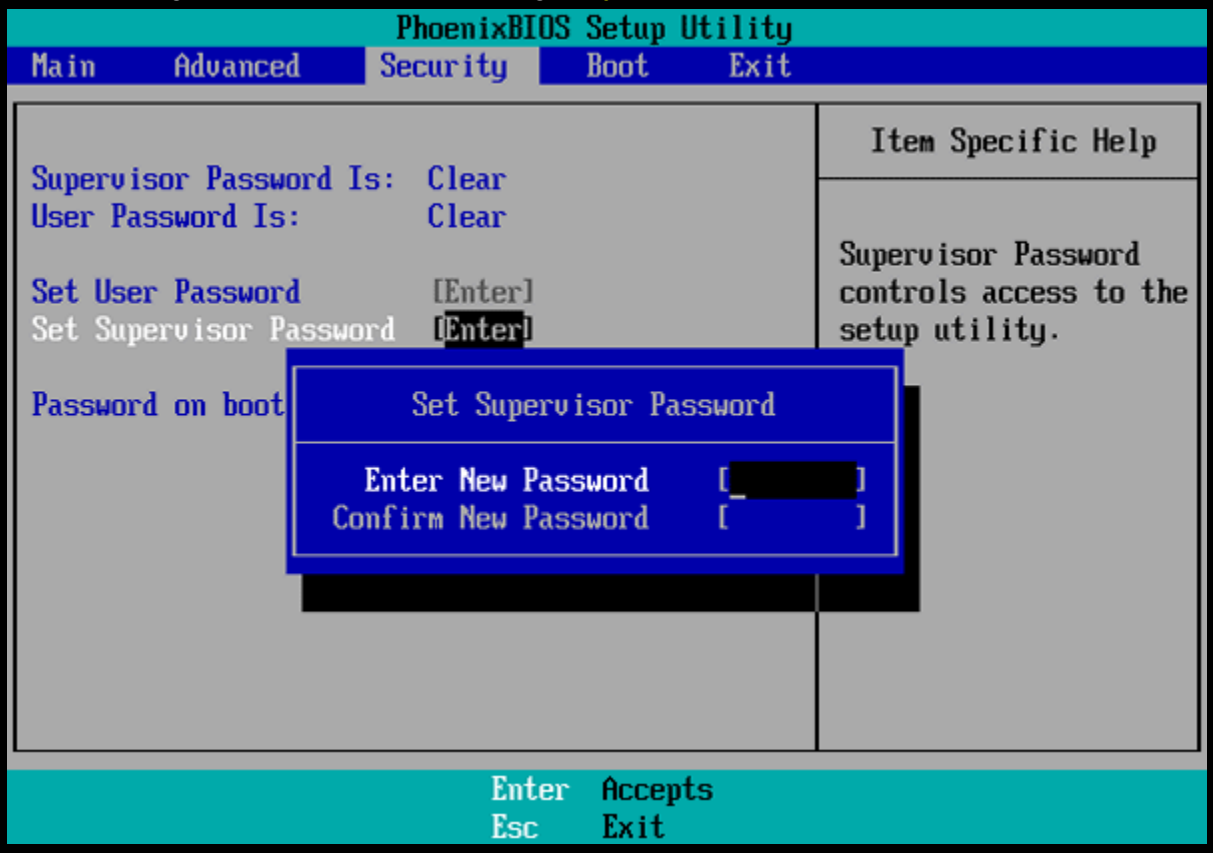
3. **XMP**→extreme memory profile→only in UEFI→only good for computers using ___DDR4_____memory→should be enabled for DDR4 memory otherwise speed is limited



4. **Temperature and Fan Settings**→found in Legacy BIOS and UEFI→can set ___temperature alarms_____ if temperature exceeds limits or fan failures



5. **Passwords→set** Administrator BIOS passwords to prevent access to the BIOS→there are also User passwords to restrict access to the computer OS boot/view the BIOS settings→these can be reset using ___jumpers_____ on the MB



All the settings can be physically cleared by removing the __CMOS battery_____.

Questions and Exercises

1. Do some research and explain the relationship between the CMOS, CMOS battery and the BIOS.

   CMOS stores the BIOS settings (time, boot order, etc.).

   BIOS is firmware stored on a chip that controls startup.

CMOS battery provides power so the CMOS settings don't get erased when the PC is off.

2. What is POST and how does it relate to the BIOS?

POST (Power-On Self Test) is the test BIOS runs at startup to check if hardware like RAM, CPU, and keyboard are working properly before loading the OS.

3. Why is the BIOS often referred to as firmware?

Because it is **software permanently stored in hardware**, not on a hard drive. It's between hardware and software → "firmware".

4. Describe the relationship of the BIOS and the MBR.

BIOS reads the **MBR** on the first bootable device.

The MBR tells BIOS **where the bootloader is**.

5. What is a bootloader and where is it commonly found?

A bootloader is a small program that loads the operating system.

It is usually found in the **MBR** or **EFI System Partition (ESP)**.

6. Find the online manual for your computer and find the specific page in the manual that states the key or key combination to access its BIOS. Enter the URL to the manual below and below the URL state the key to be pressed:

7. Try to access your BIOS using the key(s) you found in the last step. Try resetting your BIOS settings to their default or factory settings. Why would someone want to do this?

   fixing boot problems

   clearing wrong settings

   after hardware changes

   troubleshooting

8. While in your computer's BIOS find the area where you can change your boot order. Change it so that it is different from the default order. What are some of the other devices that you can set to be booted from?

   USB drive

   DVD/CD

   Network (PXE)

   External HDD

   Other internal drives

9. Research what the Fast Boot option is used for and explain it here.

Fast Boot skips some hardware checks to make the startup **faster**.

10. Which is newer: UEFI or Legacy BIOS?

    UEFI is newer.

11. What is another name used to refer to the original BIOS?

    Legacy BIOS.

12. How can you tell visually if your computer has the Legacy BIOS vs the UEFI?

    Legacy BIOS = **blue or black text-only screen**, keyboard only

    UEFI = **graphical interface**, mouse support

13. What does setting the Boot Option to start with a USB mean?

The computer will try to start an OS stored on a **USB drive** instead of the internal hard drive.

14. Under what circumstances is it important to set the Secure Boot option to enabled?Enabled when:

    you want **official Windows** protection against malware

    Disabled when:

    installing **Linux** or another unsigned OS

15. What hardware is needed in your computer to enable XMP?

    DDR4 or DDR5 RAM

    A motherboard that supports XMP

    A CPU that supports the faster RAM rates

16. How can setting a temperature alarm help you?

It warns you when the PC is overheating so you can shut it down before damaging the hardware.

17. What is the difference between a BIOS admin and user password?

    Admin password: controls access to the BIOS settings

    User password: controls access to the computer when booting

18. Go to https://www.grs-software.de/sims/bios/phoenix/pages/ and look through the example of a BIOS simulator. Is this a Legacy or UEFI BIOS?

It is **Legacy BIOS** because it uses a text-based interface.

19. What setting allows you to set the Admin password?

    Usually under:

    Security → Administrator Password

20. What is the first device to boot from?

Usually: **Hard Drive** or **PCI HDD**

21. Go to https://download.lenovo.com/bsco/index.html#/ and find a computer with a UEFI BIOS. Paste the URL to that BIOS here:

    Tell me **your Lenovo model**

22. Do the same but this time search for a computer with a Legacy BIOS:

23. Go to https://pranx.com/bios/ and determine the second device that will boot. Under what conditions would an OS be loaded from this device?

    Commonly: **CD-ROM**

    OS would load from it if the first device **has no OS**.

24. Motherboards often have jumpers (little plastic tab-like objects that fit over 2 pins that complete an electrical path between the pins) that you can use to reset the BIOS password without taking out the CMOS battery and losing all settings. Go page 32 of this manual (https://dl.dell.com/topicspdf/optiplex-3050-desktop_owners-manual3_en-us.pdf) and find the location of the jumper.

    Page 32 shows a jumper labeled:

    PSWD (Password Reset Jumper)

    Usually near the **CMOS battery**.

25. Find an image of a motherboard jumper and embed it below:

26. Read the article at https://www.electronicshub.org/how-to-identify-bios-chip-on-motherboard/ and answer the following:
    a. How can you find a BIOS chip on your motherboard?

       Look for a chip labeled **BIOS**, **UEFI**, **SPI**, or **25xx series**

       Often near the CMOS battery

       Usually 8-pin black rectangular chip

    b. What do some motherboards have two BIOS chips?

       For **backup**.

       If the main BIOS becomes corrupted, the backup BIOS can restore the system.